Live Data

This guide describes how to fetch live data of instruments easily using the SDK.

Get Quote

Fetch a real-time quote for an individual instrument using this `get_quote` method.

Python SDK Usage

```python
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

quote_response = groww.get_quote(
    exchange=groww.EXCHANGE_NSE,
    segment=groww.SEGMENT_CASH,
    trading_symbol="NIFTY"
)
print(quote_response)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange
segment *	string	Segment of the instrument such as CASH, FNO etc.

Name	Type	Description
trading_symbol *	string	Trading Symbol of the instrument as defined by the exchange

* required parameters

Response Payload

All prices in rupees.


```
{
  "average_price": 150.25,
  "bid_quantity": 1000,
  "bid_price": 150,
  "day_change": -0.5,
  "day_change_perc": -0.33,
  "upper_circuit_limit": 151,
  "lower_circuit_limit": 148.5,
  "ohlc": {
    "open": 149.50,
    "high": 150.50,
    "low": 148.50,
    "close": 149.50
  },
  "depth": {
    "buy": [
      {"price": 100.5, "quantity": 1000}
    ],
    "sell": [
      {"price": 100.5, "quantity": 1000}
    ]
  },
  "high_trade_range": 150.5,
  "implied_volatility": 0.25,
  "last_trade_quantity": 500,
  "last_trade_time": 1633072800000,
  "low_trade_range": 148.25,
  "last_price": 149.5,
  "market_cap": 5000000000,
```


Response Schema

Name	Type	Description
average_price	float	Average price of the instrument
bid_quantity	int	Quantity of the bid
bid_price	float	Price of the bid
day_change	float	Day change in price
day_change_perc	float	Day change percentage
upper_circuit_limit	float	High price range
lower_circuit_limit	float	Low price range
open	float	Opening price
high	float	Highest price
low	float	Lowest price
close	float	Closing price
price	float	Price of the book entry
quantity	int	Quantity of the book entry
high_trade_range	float	High trade range
implied_volatility	float	Implied volatility
last_trade_quantity	int	Last trade quantity
last_trade_time	int	Last trade time in epoch milliseconds
low_trade_range	float	Low trade range
last_price	float	Last traded price
market_cap	float	Market capitalization

Name	Type	Description
offer_price	float	Offer price
offer_quantity	int	Quantity of the offer
oi_day_change	float	Open interest day change
oi_day_change_percentage	float	Open interest day change percentage
open_interest	float	Open interest
previous_open_interest	float	Previous open interest
total_buy_quantity	float	Total buy quantity
total_sell_quantity	float	Total sell quantity
volume	int	Volume of trades
week_52_high	float	52-week high price
week_52_low	float	52-week low price

Get LTP

Fetch the last traded price for multiple instruments using this `get_ltp` method.

Upto 50 instruments are supported for each function call.

Python SDK Usage

```
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
```

```python
groww = GrowwAPI(API_AUTH_TOKEN)

ltp_response = groww.get_ltp(
  segment=groww.SEGMENT_CASH,
  exchange_trading_symbols="NSE_NIFTY"
)
print(ltp_response)

# you can pass multiple instruments at once
multiple_ltp_response = groww.get_ltp(
  segment=groww.SEGMENT_CASH,
  exchange_trading_symbols=("NSE_NIFTY", "NSE_RELIANCE")
)
print(multiple_ltp_response)
```

Request Schema

Name	Type	Description
segment *	string	Segment of the instrument such as CASH, FNO etc.
exchange_trading_symbols *	Tuple[str]	String of trading symbols with their respective exchanges

* required parameters

Response Payload

All prices in rupees.

```python
{
  "NSE_RELIANCE": 2500.5,
  "NSE_NIFTY": 22962.10
}
```

Response Schema

Name	Type	Description
ltp	float	Last traded price

Get OHLC

Get the OHLC details for list of given instruments quickly using this `get_ohlc` method. Use the segment value FNO for derivatives and CASH for stocks and index.

Upto 50 instruments are supported for each function call.

Note: The OHLC data retrieved using the OHLC SDK method reflects the current time's OHLC (i.e., real-time snapshot). For interval-based OHLC data (e.g., 1-minute, 5-minute candles), please refer to the Historical Data SDK method.

Python SDK Usage

```
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

ohlc_response = groww.get_ohlc(
  segment=groww.SEGMENT_CASH,
  exchange_trading_symbols="NSE_NIFTY"
)
print(ohlc_response)

# you can pass multiple instruments at once
multiple_ohlc_response = groww.get_ohlc(
  segment=groww.SEGMENT_CASH,
  exchange_trading_symbols=("NSE_NIFTY", "NSE_RELIANCE")
)
print(multiple_ohlc_response)
```

Request Schema

Name	Type	Description
segment *	string	Segment of the instrument such as CASH, FNO etc.
exchange_trading_symbols *	Tuple[str]	String of trading symbols with their respective exchanges

* required parameters

Response Payload

All prices in rupees.

```
{
  "NSE_NIFTY": {
    "open": 22516.45,
    "high": 22613.3,
    "low": 22526.4,
    "close": 22547.55
  },
  "NSE_RELIANCE": {
    "open": 1212.8,
    "high": 1215.0,
    "low": 1201.0,
    "close": 1204.0
  }
}
```

Response Schema

Name	Type	Description
open	float	Opening price
high	float	Highest price
low	float	Lowest price

Name	Type	Description
close	float	Closing price

Get Option Chain

Fetch the complete option chain data for FNO (Futures and Options) contracts including Greeks using this `get_option_chain` method.

Option chains are lists of available contracts for a specific underlying symbol and expiry date. This method provides comprehensive data including Greeks, LTP, open interest, and volume for all available strikes for both Call (CE) and Put (PE) options.

Python SDK Usage

```
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

# Get option chain for NIFTY with specific expiry date
option_chain_response = groww.get_option_chain(
    exchange=groww.EXCHANGE_NSE,
    underlying="NIFTY",
    expiry_date="2025-11-28"
)
print(option_chain_response)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange - NSE or BSE

Name	Type	Description
underlying *	string	Underlying symbol for the contract such as NIFTY, BANKNIFTY, RELIANCE etc.
expiry_date *	string	Expiry date of the contract in YYYY-MM-DD format.

* required parameters

Response Payload

All prices in rupees.


```
{
  "underlying_ltp": 25641.7,
  "strikes": {
    "23400": {
      "CE": {
        "greeks": {
          "delta": 0.9936,
          "gamma": 0,
          "theta": -1.0787,
          "vega": 0.6943,
          "rho": 5.1802,
          "iv": 25.3409
        },
        "trading_symbol": "NIFTY25N1823400CE",
        "ltp": 2200,
        "open_interest": 7,
        "volume": 5
      },
      "PE": {
        "greeks": {
          "delta": -0.0064,
          "gamma": 0,
          "theta": -1.0787,
          "vega": 0.6943,
          "rho": -0.0373,
          "iv": 25.3409
        },
        "trading_symbol": "NIFTY25N1823400PE",
        "ltp": 2.05,
```

Response Schema

Name	Type	Description
underlying_ltp	float	Last traded price of the underlying instrument
strikes	object	Strike-wise option chain data containing CE and PE contracts
trading_symbol	string	Trading symbol of the option contract
ltp	float	Last traded price of the option contract
open_interest	int	Open interest for the contract
volume	int	Total volume traded for the contract
delta	float	Delta measures the rate of change of option price based on every 1 rupee change in the price of underlying
gamma	float	Gamma measures the rate of change of delta with respect to underlying asset price
theta	float	Theta measures the rate of time decay of option price
vega	float	Vega measures the rate of change of option price based on every 1% change in implied volatility
rho	float	Rho measures the sensitivity of option price to changes in interest rates
iv	float	Implied Volatility represents the market's expectation of future volatility, expressed as a percentage

Get Greeks

Fetch the Greeks data for FNO (Futures and Options) contracts using this `get_greeks` method.

Greeks are financial measures that help assess the risk and sensitivity of options contracts to various factors like underlying price changes, time decay, volatility, and interest rates. This method is

specifically designed for derivatives trading and risk management.

Python SDK Usage

```python
from growwapi import GrowwAPI

# Groww API Credentials (Replace with your actual credentials)
API_AUTH_TOKEN = "your_token"

# Initialize Groww API
groww = GrowwAPI(API_AUTH_TOKEN)

greeks_response = groww.get_greeks(
    exchange=groww.EXCHANGE_NSE,
    underlying="NIFTY",
    trading_symbol="NIFTY25O1425100CE",
    expiry="2025-10-14"
)
print(greeks_response)
```

Request Schema

Name	Type	Description
exchange *	string	Stock exchange - NSE or BSE
underlying *	string	Underlying symbol for the contract such as NIFTY, BANKNIFTY, RELIANCE etc.
trading_symbol *	string	Trading Symbol of the FNO contract as defined by the exchange
expiry *	string	Expiry date of the contract in YYYY-MM-DD format

* required parameters

Response Payload

```
{
  "greeks": {
```

```
    "delta": 0.6006,
    "gamma": 0.0014,
    "theta": -8.1073,
    "vega": 13.1433,
    "rho": 2.7333,
    "iv": 8.2383
  }
}
```

Response Schema

Name	Type	Description
delta	float	Delta measures the rate of change of option price based on every 1 rupee change in the price of underlying.
gamma	float	Gamma measures the rate of change of delta with respect to underlying asset price. Higher gamma means delta changes more rapidly
theta	float	Theta measures the rate of time decay of option price. Usually negative, indicating option value decreases over time
vega	float	Vega measures the rate of change of option price based on every 1% change in implied volatility of the underlying asset
rho	float	Rho measures the sensitivity of option price to changes in interest rates
iv	float	Implied Volatility represents the market's expectation of future volatility, expressed as a percentage

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